

Robustness Checks for Acs (2025): Mapping the Political Contours of the Regulatory State: Dynamic Estimates of Agency Ideal Points

PLSC 503: Homework 3

Jody Park

2026-04-20

1. Replication Setup

Citations and Links

- **Original Paper:** ACS, A. (2025). Mapping the Political Contours of the Regulatory State: Dynamic Estimates of Agency Ideal Points. *American Political Science Review*, 1-18.
- **Replication Data:** [R replication codes and dataset](#) from Acs (2025) “Mapping the Political Contours of the Regulatory State: Dynamic Estimates of Agency Ideal Points (APSR).”
- **Paper for the Extension:** Fazekas, M., Ferrali, R., & Wachs, J. (2023). Agency Independence, Campaign Contributions, and Favoritism in US Federal Government Contracting. *Journal of Public Administration Research and Theory*, 33(2), 262-278.
- **Replication Data:** [R replication codes and dataset](#) from Fazekas et al. (2023) “Agency Independence, Campaign Contributions, and Favoritism in US Federal Government Contracting (JPART).”
- **Repository Links:**
 - [Dropbox folder](#); current code and data for this paper are in the *acs-2025-apsr* subfolder.
 - [GitHub repository for this project](#)

Article Introduction

Acs (2025) investigates the political ideology of US federal regulatory agencies by estimating dynamic ideal points for 69 administrative units across four presidential administrations (Clinton to Trump). The study uses data from the Unified Agenda on planned regulations and the president’s discretionary review of those regulations from the Office of Information and Regulatory Affairs (OIRA) docket. By analyzing the ideological positions of regulatory agencies over time, the paper provides insights into how agency ideology shapes regulatory behavior and how it interacts with political dynamics. The core contribution of the paper is to offer a comprehensive mapping of the political contours of the regulatory state, shedding light on the ideological underpinnings of regulatory activity and its implications for policy outcomes.

I extend the analysis by examining how agency ideology moderates the effects of political party contributions on favoritism in federal contracting, using the dynamic ideal point estimates from Acs (2025). Merging this agency ideology data with Fazekas et al. (2023) allows a direct test of whether ideological alignment between the president and an agency amplifies the returns to political contributions in procurement. For this homework, I conduct three robustness checks to evaluate how much confidence the matched procurement sample can support: (1) a bootstrap to assess small-sample uncertainty, (2) a leave-one-agency-out sensitivity analysis to identify influential cases, and (3) a power-style simulation to evaluate the likelihood of detecting

true effects given the sample size and effect sizes observed in the matched panel. These checks help determine whether the procurement results are precise enough to support substantive claims about the role of agency ideology in shaping procurement outcomes.

Data, Variables and Models

Data

In Acs (2025), the author uses publicly available datasets:

1. **Unified Agenda data:** Includes information on the regulations that federal agencies plan to implement, including their content, timing, and regulatory impact (a semi-annual catalog, indicating pending or completed).
2. **The Office of Information and Regulatory Affairs (OIRA) docket:** Lists the regulations reviewed. For each regulation, the docket includes information on the agency responsible, the date of review, and the outcome of the review (e.g., approved, disapproved, or returned for revision).
3. **Federal Register data:** Contains the official publication of federal regulations and related documents. [Federal Register API](#)

Acs (2025) uses the eight-digit regulation identifier number (the RIN) to merge the Unified Agenda data with the OIRA docket data.

- The analysis focuses on 69 administrative units (e.g., EPA, FDA, etc.) from 2001 to 2020, covering four presidential administrations (Clinton, Bush, Obama, Trump).
 - The author excludes regulatory agencies that “did not issue a regulation in each administration or issued fewer than 40 regulations during the entire period (averaging less than 10 per administration)” (Acs, 2025).
 - This exclusion criterion ensures that the analysis focuses on regulatory agencies with sufficient regulatory activity to estimate meaningful ideal points and examine their dynamics over time.

Variables

1. Dependent Variable:

- (1) **Review (Y_i)** (for the dynamic measurement model): Binary indicator for whether a planned rule in the Unified Agenda was reviewed by OIRA (1 = reviewed at least once in the OIRA docket; 0 = not reviewed).
- (2) **Regulatory Activity $_{\{at\}}$** (for regulatory productivity model): Agency-level regulatory output in administration t , measured using Federal Register final-rule page length (average per year within administration, with alternative coding rules reported in Table 2 and Supplementary Appendix).

2. Independent Variable:

- **President-Agency ideological distance:** $|X_p - X_{\{at\}}|$, where X_p is the president’s fixed ideal point (DW-NOMINATE anchors) and $X_{\{at\}}$ is the agency ideal point in administration t .

3. Model Components:

- **Rule significance (S_i):** Binary indicator from Unified Agenda priority categories (1 = economically significant/significant; 0 = otherwise), plus interaction with spatial distance in the measurement model.
- **Model structure:** **Agency intercepts (a_a)** and **dynamic structure:** Agency-specific baseline review propensity and a random-walk evolution of agency ideal points across administrations (Clinton, Bush, Obama, Trump), estimated in Stan (Supplementary Appendix B).

Models

For this homework, I will focus on *Model 2* (Regulatory Productivity) and *H3*, which use a negative binomial model to examine the relationship between President-Agency ideological distance and regulatory output.

- **Model 1 (Dynamic Measurement Model / IRT-Probit):** The core model estimates agency ideal points from OIRA review decisions using a Bayesian probit IRT structure with president-agency distance, significance terms, agency intercepts, and administration-specific agency ideal points that evolve via a random walk. The Stan specification and priors are provided in Supplementary Appendix B.
- **Model 2 (Validation Model: Regulatory Productivity):** The paper then estimates a *negative binomial model* (Equation 3 in the original paper) where regulatory activity is regressed on President-Agency Distance with agency fixed effects and multiple outcome constructions (Table 2), testing whether ideologically closer agencies are more productive.

$$\begin{aligned} \text{Regulatory Activity}_{at} &\sim \text{NegBin}(\mu_{at}, \theta) \\ \ln(\mu_{at}) &= \pi_a + \beta \cdot \text{PresidentAgencyDistance}_{at} \end{aligned} \tag{3}$$

Tested Hypotheses in the Original Paper:

- **H1 (Review-distance expectation):** OIRA review probability increases as ideological distance between the president and an agency increases.
- **H2 (Dynamic ideology expectation):** Agency ideology is not static; agency ideal points shift across presidential administrations.
- **H3 (Productivity-distance expectation):** Regulatory productivity is lower when President-Agency ideological distance is larger (negative association in Equation 3 / Table 2).

2. Replication Accuracy

Figure 4. Conservative Trends in Agency Ideal Points

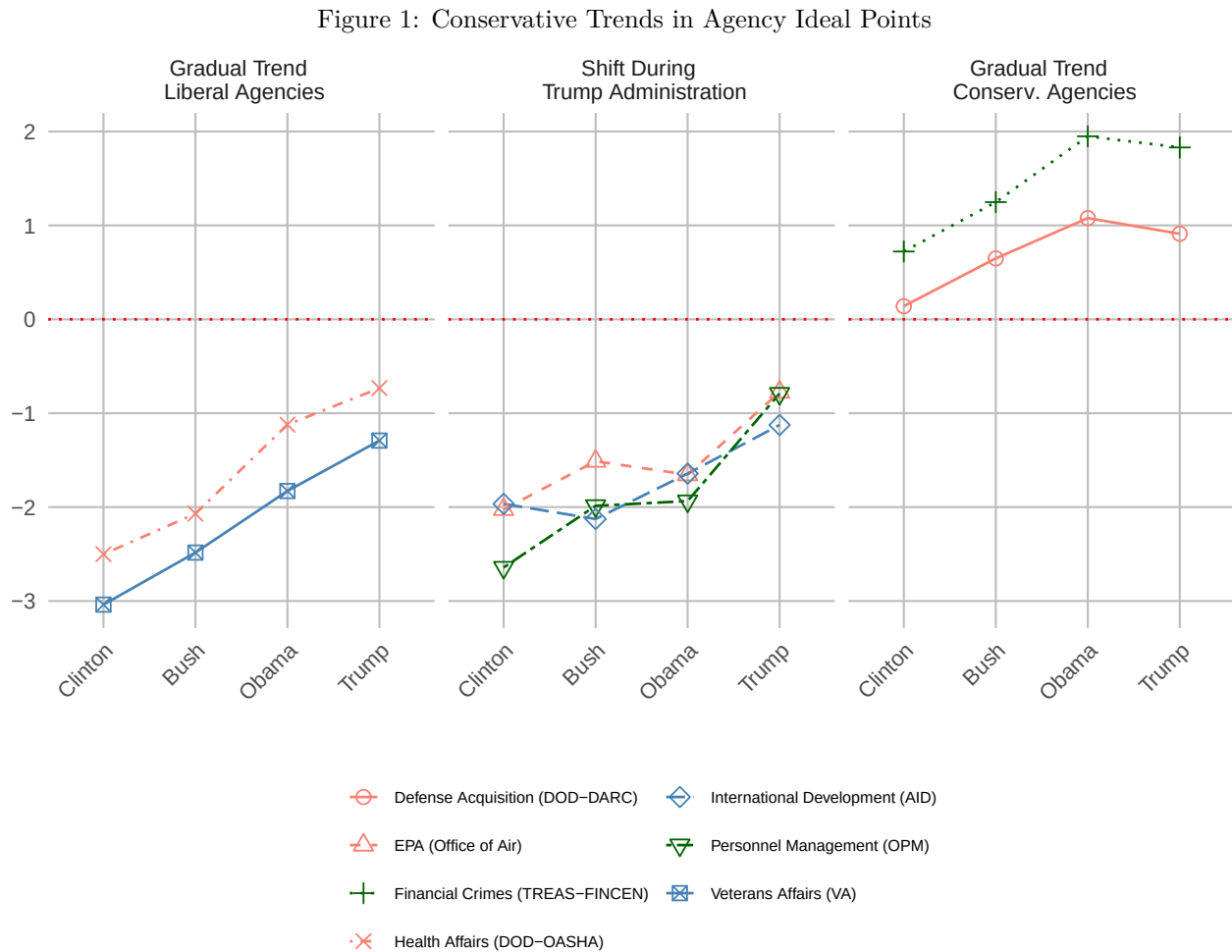


Figure 1 replicates the original pattern that several agencies shift in a conservative direction over administrations, with the largest movement concentrated in a small subset of agencies. This visual replication is consistent with Acs (2025)'s Figure 4.

Table 1. Agencies with Largest Shifts in the Conservative Direction

Table 1: Agencies with Largest Shifts in the Conservative Direction

Agency	Ideal Points		Plausible Driver of Change
	Min	Max	
Personnel Management (OPM)	-2.64	-0.79	<i>Politicization:</i> Trump administration
EPA’s Office of Air (EPA-AIR)	-2.02	-0.78	Trump administration
Agency for Intl. Development (USAID)	-2.13	-1.13	Trump administration
Veterans Affairs (VA)	-3.04	-1.29	<i>New policy conflict:</i> VA MISSION Act of 2018, wars
Defense Acquisition (DOD-DARC)	0.14	1.08	Iraq/Afghan wars
Financial Crimes (TREAS-FINCEN)	0.72	1.95	Patriot Act of 2001, civil liberties
Health Affairs, Def. (DOD-OASHA)	-2.50	-0.73	Iraq/Afghan wars

Table 1 reproduces the agency-level minima and maxima from the Figure 4 subset and highlights the same conservative movers. The ordering and substantive interpretation follow the original table.

Table 2. Ideological Distance and Regulatory Productivity (Negative Binomial)

Table 2: Ideological Distance and Regulatory Productivity (Negative Binomial)

	Dependent variable: Regulatory activity (final rule page length)				
	(1)	(2)	(3)	(4)	(5)
	Formal	Formal significant	Formal significant regulatory	Formal (excl. OIRA)	Formal regulatory (excl. OIRA)
Pres.-Agency Distance	-0.048 (0.036)	-0.186* (0.049)	-0.218* (0.054)	-0.152* (0.036)	-0.218* (0.038)
Agency Fixed Effects	✓	✓	✓	✓	✓
Observations	276	276	276	276	276

* $p < 0.05$. Standard errors in parentheses. Models 4-5 exclude rules reviewed by OIRA at the proposed rule stage. Models 3 and 5 also exclude deregulatory rules. Weights equal to the inverse variance of each agency ideal point are included.

Table 2 replicates the negative-binomial validation result: larger President-Agency ideological distance is associated with lower regulatory productivity across outcome definitions. The sign and substantive magnitude patterns are aligned with the original paper.

Table A1. Agency Names and Ideal Point Estimates

Table 3: Agency Names and Ideal Point Estimates

agency id	clinton sig mean	clinton review mean	clinton n	bush43 sig mean	bush43 review mean	bush43 n	obama sig mean	obama review mean	obama n	trump sig mean	trump review mean	trump n
AG-AMS	0.229	0.200	35	0.125	0.125	64	0.286	0.167	42	0.222	0.222	18
AG-APHIS	0.826	0.783	46	0.676	0.648	71	0.394	0.394	66	0.150	0.150	20
AG-FNS	0.586	0.586	58	0.857	0.821	28	0.390	0.439	41	0.692	0.769	13
AG-FS	0.217	0.304	23	0.500	0.412	34	0.375	0.375	16	0.625	0.500	8
AG-FSIS	0.543	0.478	46	0.375	0.250	24	0.458	0.458	24	0.176	0.176	17
AG-GIPSA	0.348	0.348	23	0.059	0.059	17	0.364	0.364	11	1.000	1.000	1
AG-RHS	0.450	0.450	20	0.154	0.154	13	0.167	0.083	12	0.000	0.000	9
AG-RUS	0.194	0.226	31	0.174	0.130	23	0.083	0.083	12	0.000	0.000	1
AID	0.167	0.333	6	0.667	0.889	9	0.625	0.500	16	0.444	0.444	9
COM-BEA	0.000	0.000	13	0.000	0.080	25	0.000	0.000	12	0.000	0.200	5
COM-BIS	0.800	0.800	5	0.400	0.300	20	0.721	0.744	43	0.750	0.750	8
COM-ITA	0.143	0.143	7	0.200	0.200	15	0.118	0.059	17	1.000	1.000	4
COM-NOAA	0.060	0.101	398	0.125	0.133	534	0.061	0.051	528	0.067	0.067	252
COM-PTO	0.050	0.150	20	0.289	0.316	38	0.654	0.692	26	0.214	0.357	14
DHS-USCBP	0.667	0.000	3	0.565	0.391	23	0.357	0.357	14	0.273	0.182	11
DHS-USCG	0.267	0.000	15	0.315	0.185	54	0.143	0.125	56	0.111	0.111	18
DHS-USCIS	1.000	0.000	8	1.000	0.700	30	1.000	1.000	10	0.889	0.722	18
DOJ-DARC	0.022	0.022	90	0.050	0.050	141	0.280	0.318	132	0.202	0.112	89
DOJ-DOAOASHA	0.036	0.393	28	0.484	0.839	31	0.212	0.182	33	0.125	0.125	8
DOE-EE	0.444	0.250	36	0.406	0.375	32	0.361	0.340	147	0.316	0.368	38
DOI-BIA	0.239	0.109	46	0.136	0.364	22	0.231	0.231	13	0.000	0.000	7
DOI-BLM	0.206	0.206	34	0.235	0.235	17	0.462	0.615	13	0.500	0.625	8
DOI-FWS	0.077	0.181	310	0.406	0.206	310	0.091	0.103	331	0.100	0.100	120
DOI-MMS	0.157	0.255	51	0.178	0.178	45	0.545	0.273	11	1.000	1.000	1
DOI-NPS	0.222	0.361	36	0.286	0.306	49	0.139	0.139	36	0.188	0.250	16
DOJ-BOP	0.125	0.031	32	0.750	0.700	20	0.533	0.467	15	0.333	0.333	3
DOJ-DEA	0.250	0.333	24	0.710	0.871	31	0.727	0.727	11	0.800	0.600	10
DOJ-EOIR	0.222	0.111	9	0.400	0.800	15	0.600	0.600	5	1.000	0.727	11
DOJ-LA	0.400	0.450	20	0.850	0.600	20	0.500	0.417	12	0.800	0.800	5
DOL-EBSA	0.588	0.647	17	0.850	0.900	20	0.964	0.857	28	1.000	0.833	12
DOL-ETA	1.000	1.000	9	1.000	0.889	18	1.000	0.917	12	1.000	1.000	8
DOL-MSHA	0.583	0.417	12	0.947	0.842	19	1.000	1.000	8	0.667	0.667	3
DOL-OSHA	0.773	0.273	22	0.722	0.833	18	0.435	0.304	23	0.889	0.889	9
DOL-PBGC	0.312	0.188	16	0.714	0.643	14	0.308	0.385	13	0.000	0.000	4
DOT-FAA	0.422	0.296	135	0.308	0.240	104	0.267	0.280	75	0.304	0.261	23
DOT-FHWA	0.308	0.231	78	0.289	0.289	38	0.419	0.387	31	0.417	0.250	12
DOT-FMCSA	0.611	0.139	36	0.632	0.632	19	0.464	0.393	28	0.200	0.240	25
DOT-FRA	0.652	0.478	23	0.348	0.261	23	0.342	0.316	38	0.250	0.250	16
DOT-FTA	0.389	0.167	18	0.375	0.188	16	0.500	0.450	20	0.333	0.000	3
DOT-NHTSA	0.107	0.114	140	0.228	0.228	101	0.304	0.261	69	0.267	0.333	15
DOT-OST	0.255	0.218	55	0.293	0.317	41	0.375	0.417	24	0.417	0.417	12
DOT-PRMISA	0.222	0.159	63	0.280	0.240	50	0.327	0.269	52	0.500	0.500	10
ED-OESE	0.800	0.600	5	0.889	0.889	9	0.727	0.818	22	0.000	0.000	4
ED-OPE	0.679	0.464	28	1.000	1.000	5	0.955	0.955	22	1.000	0.818	11
EPA-AR	0.361	0.354	291	0.374	0.505	281	0.525	0.506	263	0.382	0.382	123
EPA-OPPTS	0.591	0.455	44	0.333	0.381	21	0.558	0.538	52	0.500	0.607	28
EPA-SWER	0.650	0.567	60	0.771	0.686	35	0.484	0.484	31	0.882	0.765	17
EPA-WATER	0.412	0.425	80	0.362	0.468	47	0.488	0.488	41	0.706	0.647	17
FAR	0.171	0.238	181	0.124	0.114	105	0.621	0.589	95	0.317	0.366	41
GSA	0.323	0.355	62	0.125	0.083	48	0.171	0.122	41	0.667	0.533	15
HHS-ACF	0.548	0.714	42	0.833	1.000	18	0.929	0.857	14	0.857	1.000	7
HHS-CMS	0.718	0.825	103	0.897	0.960	174	0.985	0.923	196	0.978	0.944	90
HHS-FDA	0.620	0.479	121	0.726	0.500	62	0.618	0.500	68	0.500	0.525	40
HUD-OH	0.548	0.595	42	0.821	0.821	56	0.440	0.460	50	0.625	0.625	8
HUD-PHI	0.949	0.846	39	0.867	0.833	30	0.706	0.588	17	1.000	1.000	5
NARA	0.389	0.389	18	0.194	0.258	31	0.625	0.562	16	0.333	0.333	3
NASA	0.111	0.111	9	0.059	0.059	17	0.364	0.182	33	0.333	0.333	6
NCUA	0.000	0.000	31	0.014	0.000	72	0.046	0.000	65	0.000	0.000	38
OPM	0.136	0.544	103	0.558	0.865	104	0.528	0.613	106	0.684	0.447	38
RRB	0.097	0.097	31	0.385	0.385	13	0.200	0.000	5	0.000	0.000	2
SBA	0.627	0.520	75	0.784	0.745	51	0.614	0.649	57	0.385	0.269	26
SSA	0.439	0.316	57	0.775	0.800	80	0.895	0.684	38	0.800	0.700	10
STATE	0.111	0.333	9	0.405	0.514	37	0.776	0.690	58	0.857	0.857	21
TREAS-CUSTOMS	0.015	0.000	66	0.000	0.000	22	0.278	0.222	18	0.714	0.571	7
TREAS-DO	0.167	0.000	12	0.275	0.250	40	0.571	0.571	21	0.400	0.100	10
TREAS-FINCEN	0.000	0.000	11	0.062	0.031	32	0.630	0.593	27	0.750	0.250	4
TREAS-OCC	0.023	0.047	43	0.278	0.083	36	0.256	0.070	43	0.179	0.000	39
TREAS-TTB	0.000	0.000	7	0.055	0.018	55	0.050	0.050	20	0.500	0.500	2
VA	0.228	0.596	114	0.580	0.902	112	0.409	0.299	127	0.343	0.388	67

Table 3 reproduces the appendix-style agency-by-administration summaries of significance rates, review rates, and counts. These summaries support the data pipeline used to estimate the main replication models.

3. Methodological Extension

Why this method is necessary

Section 2 shows that I can reproduce the main ACS validation results with the original agency-administration panel. My [HW2 extension](#), however, depends on a much smaller matched ACS-Fazekas panel. The ACS replication in Section 2 uses an agency-administration panel with $N = 276$, which is large enough to estimate the negative binomial validation model in [Acs \(2025\)](#) with relatively stable coefficients. Once I merge ACS ideological distance with the contracting data, the effective sample size falls to 48 agency-administration observations in the main procurement models, 42 in the ordered logit, and 22 in the agency-level follow-up. The matched procurement panel contains 25 distinct agencies, and only 22 remain once the agency-independence categories needed for the ordered-logit checks are imposed.

That drop in sample size creates a different inferential problem than the one emphasized in the original ACS article. The concern is no longer whether the model form is correct. Instead, the concern is whether coefficient instability reflects substantive heterogeneity or simply limited information. With a sample this small, null or mixed findings may be low-power results rather than strong evidence against the extension.

How the method improves the paper

For HW3, I keep the HW2 extension design and reframe the robustness question around sample size and power. I use three checks. First, I bootstrap the key `distance` coefficients in the four procurement models to quantify how much small-sample uncertainty widens the confidence intervals beyond the model-based standard errors. Second, I conduct a leave-one-agency-out sensitivity analysis to determine whether the results are driven by a handful of matched agencies. Third, I run a power-style simulation that projects rejection rates as the effective sample size increases from the current matched panel toward the size of the original ACS replication sample.

These checks do not replace the extension models. They evaluate whether the extension results are fragile because the matched panel is small. This is suitable for HW3 because the sample size problem is already visible in the current write-up, and because the assignment specifically asks for a robustness check that addresses sample size and power.

Robustness Check 1: Small-Sample Uncertainty via Bootstrap

The first robustness check resamples agencies with replacement and refits the four procurement models from HW2. This is an agency bootstrap rather than a row bootstrap because the two administration rows for a given agency are not substantively independent. Resampling at the agency level preserves that within-agency structure while still quantifying how much the `distance` coefficient varies under repeated sampling from a small matched panel.

If the bootstrap intervals are much wider than the asymptotic intervals from the original models, then the extension results are more fragile than the default standard errors suggest. If the sign remains stable and the bootstrap interval still excludes zero, then the small sample is less threatening to the substantive conclusion.

Robustness Check 2: Leave-One-Agency-Out Sensitivity

The second robustness check drops one matched agency at a time and re-estimates the four procurement models. This directly tests whether the extension is driven by a small number of influential agencies. With only 25 matched agencies, a single omitted case can shift the point estimate in a meaningful way.

The substantive question is: do the coefficients keep the same sign and roughly the same magnitude when each agency is omitted in turn? If the answer is yes, then the extension is not being carried by one or two special cases. If the sign flips or the coefficient collapses toward zero under many omissions, then the extension should be interpreted more cautiously.

Robustness Check 3: Power-Style Simulation

The third robustness check asks whether the current matched panel is large enough to detect effects of the size observed in HW2. I treat the estimated coefficient and standard error from each procurement model as the starting point for a retrospective design analysis. Then I project what power would look like at the current extension sample size ($N = 48$), a doubled sample ($N = 96$), a four-times larger sample ($N = 192$), and the original ACS replication size ($N = 276$).

I report power under two assumptions: the observed effect size from the HW2 extension, and a more conservative scenario in which the true effect is only half as large. This check follows the sample-size logic from class: if power remains low even under optimistic assumptions, then coefficient instability is more likely to reflect limited information than a well-estimated null effect.

Explanatory Table

Explanatory Table 4. Effective Sample Sizes Across ACS Replication and Extension Models

Table 4: Effective sample sizes across ACS replication and extension models

Model	Unit	Num.Obs.	Distinct agencies
ACS replication Table 2	Agency-administration	276	69
HW3 matched procurement panel	Agency-administration	48	25
Ordered logit extension	Agency-administration	42	22
Agency-level ordered logit	Agency	22	22

Table 4 shows the core inferential problem motivating HW3. The original ACS replication uses 276 agency-administration observations, but the matched procurement extension falls to 48 rows, and the ordered-logit follow-up models shrink further to 42 and 22. This large drop in effective sample size is the basis for the robustness checks below.

Robustness Check 1. Small-Sample Uncertainty via Bootstrap

Table 5: Bootstrap uncertainty for procurement extension coefficients

Model	Estimate	Asymptotic 95% CI	Bootstrap 95% CI	Bootstrap SD	Usable reps
Donor contract count	0.168	[-0.093, 0.444]	[-0.034, 0.309]	0.088	199
Donor contract share	-0.000	[-0.039, 0.038]	[-0.043, 0.040]	0.024	199
Log donation amount	-2.256	[-3.845, -0.667]	[-4.937, 1.189]	1.703	199
Total contracts	-0.658	[-1.285, 0.132]	[-1.957, 0.038]	0.472	199

Table 5 compares the model-based and bootstrap uncertainty around the **distance** coefficient in the four procurement models. If the bootstrap intervals are materially wider than the asymptotic intervals, then the matched extension sample is not large enough for the default standard errors to tell the full uncertainty story.

Figure 2: Asymptotic versus bootstrap confidence intervals for distance coefficients

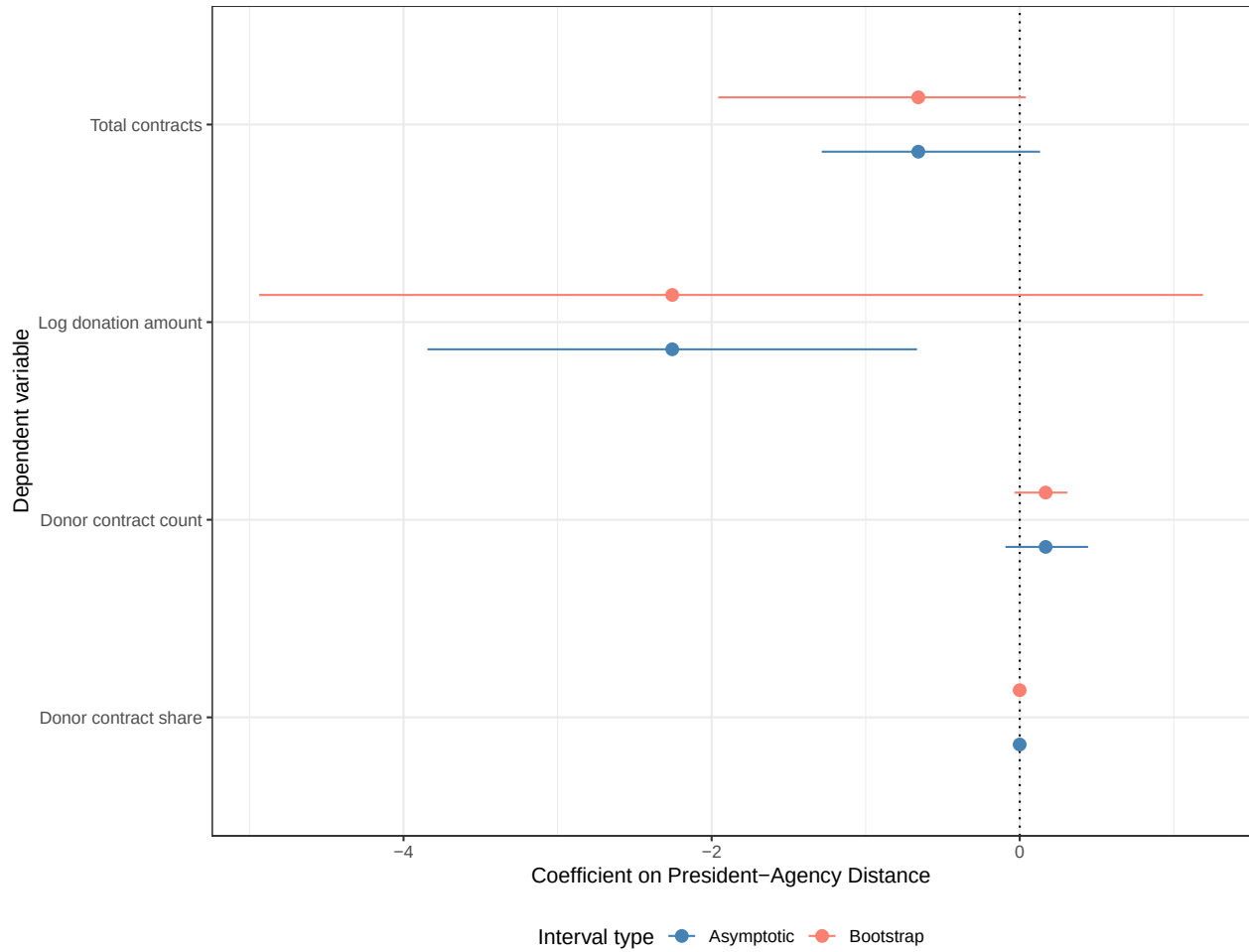


Figure 2 visualizes the same comparison. The distance coefficient is most stable when both the asymptotic and bootstrap intervals remain on the same side of zero. Wider bootstrap intervals indicate that the small matched sample adds uncertainty not obvious from the default model output alone.

Robustness Check 2. Leave-One-Agency-Out Sensitivity

Table 6: Leave-one-agency-out sensitivity summary

model	Full estimate	LOO range	LOO SD	Share same sign
Donor contract count	0.168	[0.087, 0.207]	0.021	1.00
Donor contract share	-0.000	[-0.014, 0.027]	0.007	0.72
Log donation amount	-2.256	[-3.343, -0.330]	0.478	1.00
Total contracts	-0.658	[-0.916, -0.402]	0.092	1.00

Table 6 summarizes how much the **distance** coefficient changes when one agency is omitted at a time. The most reassuring case is when the range is narrow and the sign remains the same across nearly all leave-one-out estimates. Wider ranges and sign reversals indicate that the result is being carried by a small number of agencies.

Figure 3: Leave-one-agency-out dispersion in distance coefficients

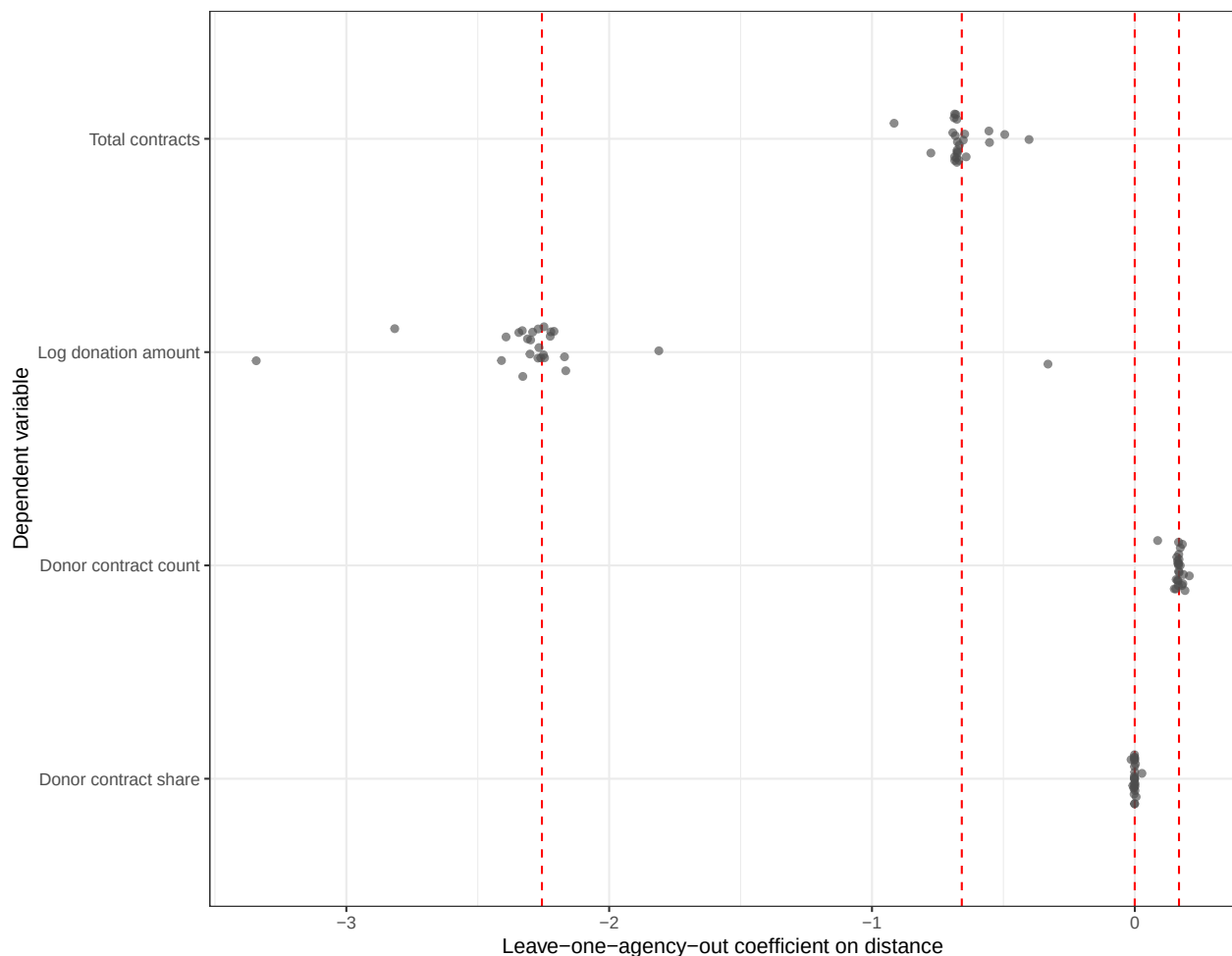


Figure 3 shows the full distribution of leave-one-agency-out estimates. The dashed red line is the full-sample coefficient. Tight clusters around that line indicate stable results. Wide clouds or points crossing zero indicate that the small matched panel is sensitive to which agencies are included.

Robustness Check 3. Power-Style Simulation

Table 7: Power-style simulation for the matched procurement panel

Model	Assumption	Power	Type S	Type M
Donor contract count	Half observed effect	0.096	0.050	3.962
Donor contract count	Observed effect	0.245	0.002	2.066
Donor contract share	Half observed effect	0.049	0.475	212.234
Donor contract share	Observed effect	0.048	0.444	105.416
Log donation amount	Half observed effect	0.295	0.001	1.809
Log donation amount	Observed effect	0.816	0.000	1.110
Total contracts	Half observed effect	0.181	0.008	2.540
Total contracts	Observed effect	0.497	0.000	1.393

Table 7 reports the power-style simulation at the current matched sample size. Power is shown under two assumptions: the observed HW2 effect size and a more conservative half-effect scenario. Type S is the probability of estimating the wrong sign conditional on statistical significance, and Type M is the exaggeration ratio conditional on significance. High Type M values indicate that significant results from a small sample are likely to overstate the true effect.

Figure 4: Projected power as the matched sample size grows

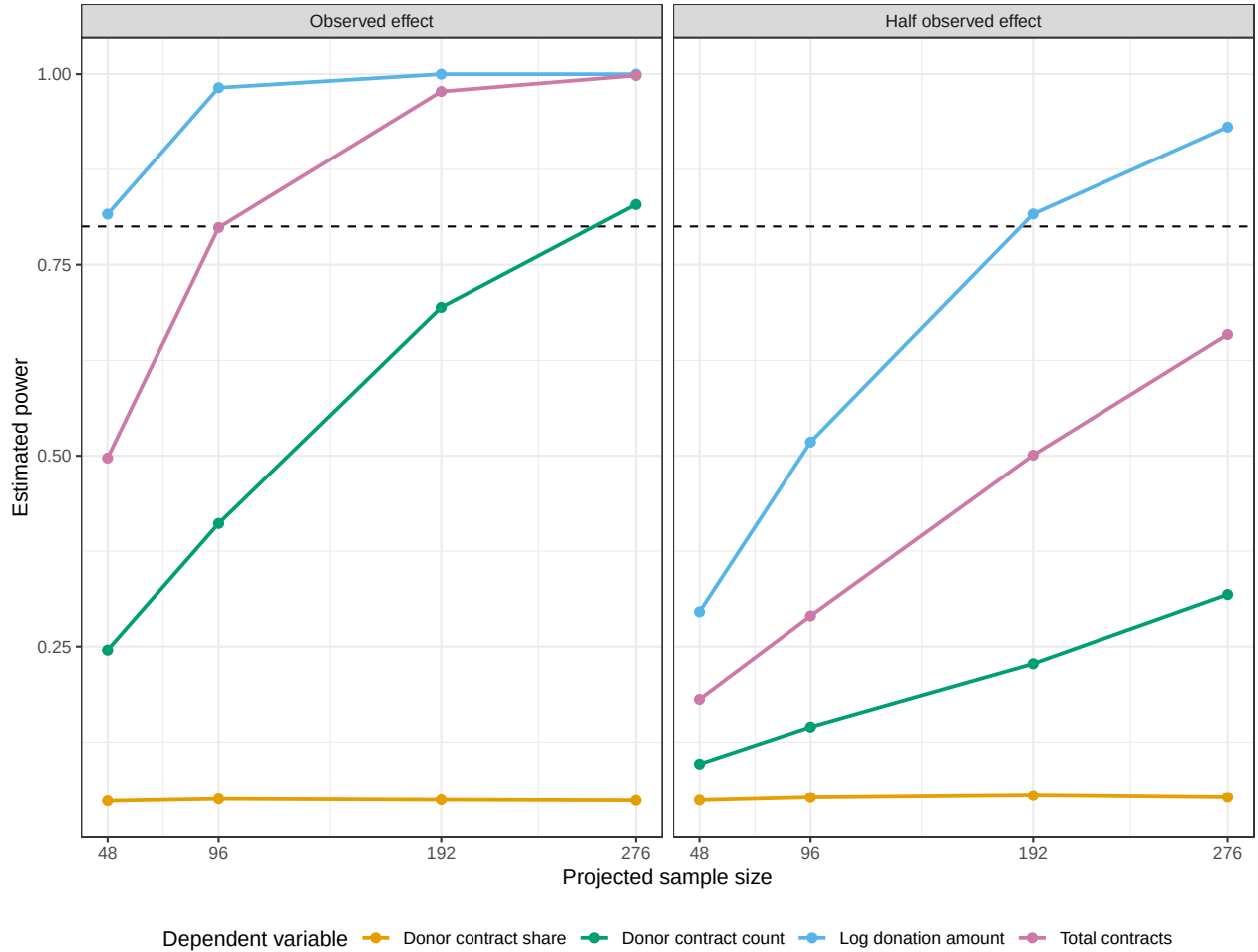


Figure 4 traces how power changes as the matched panel grows. The dashed horizontal line marks the conventional 0.80 benchmark. If the lines remain well below 0.80 at $N = 48$, then the current extension is underpowered even before considering model uncertainty from agency matching.

4. Results

Table 4. Effective Sample Sizes Across ACS Replication and Extension Models

Table 4 shows the sample-size contrast motivating the HW3 robustness design. The original ACS replication uses 276 agency-administration observations, but the procurement extension relies on only 48 matched rows, while the ordered-logit follow-ups fall to 42 and 22 observations. This confirms that the main risk in the extension is not data abundance but limited information.

Table 5. Bootstrap Uncertainty for Procurement Extension Coefficients

Table 5 shows that the bootstrap check does not affect the four procurement models in the same way. The donor-contract-share model remains centered at zero under both interval estimates, so this outcome provides little evidence of any distance effect. The donor-contract-count model stays positive, but both the asymptotic and bootstrap intervals include zero, which suggests directional stability without much precision. The clearest change appears in the log-donation model: the asymptotic interval excludes zero, but the bootstrap interval widens enough to include zero. That shift indicates that the negative HW2 estimate is substantively interesting but less certain once small-sample uncertainty is taken seriously. The total-contracts model stays negative, yet both intervals still include zero.

Substantively, the matched sample gives the clearest support for a negative relationship between ideological distance and the total amount donated within agency-administration cells. By contrast, the extension provides much weaker evidence that ideological distance changes donor-linked contract share, donor-linked contract count, or total contracting volume with comparable precision.

Figure 2. Asymptotic versus Bootstrap Confidence Intervals

Figure 2 makes the same pattern easier to see. The bootstrap intervals for donor-contract share and donor-contract count are close to the asymptotic ones, but the log-donation and total-contract intervals widen noticeably. The log-donation result is still centered on a large negative coefficient, yet the bootstrap figure shows why that estimate should be interpreted as suggestive rather than definitive in a sample of 48 matched observations.

Table 6. Leave-One-Agency-Out Sensitivity Summary

Table 6 shows that the leave-one-agency-out story is more reassuring than the bootstrap story for three of the four models. Donor-contract count stays positive in every omission, with a narrow range from 0.087 to 0.207. Total contracts also remains negative in every omission, and its range is fairly tight. The log-donation model keeps a negative sign throughout as well, although its range is wider, which means the coefficient is not driven by a single agency but its magnitude is sensitive to which agencies are in the sample. The donor-contract-share model is the weakest case. Its full-sample coefficient is essentially zero, and the sign changes in about 28 percent of the leave-one-out estimates.

Figure 3. Leave-One-Agency-Out Dispersion in Distance Coefficients

Figure 3 reinforces that reading. The donor-contract-count and total-contract clouds are tightly clustered around the full-sample coefficient. The log-donation estimates are more dispersed and include a few visibly influential omissions, but they do not cross zero. By contrast, the donor-contract-share estimates sit almost directly on zero and move across it repeatedly, which is consistent with the absence of a stable signal in that model.

Table 7. Power-Style Simulation for the Matched Procurement Panel

Table 7 shows that the current matched panel is not equally informative across outcomes. Under the observed-effect scenario, only the log-donation model clears the conventional 0.80 power threshold at $N = 48$. Total contracts reaches only moderate power (0.497), and donor-contract count remains low (0.245).

Under the half-effect scenario, power drops sharply for every model. The donor-contract-share model is especially uninformative: because the observed coefficient is essentially zero, projected power stays at the nominal false-positive rate and the Type S and Type M summaries become mechanically unhelpful. The more informative comparison is between log donation, which remains relatively strong, and the count-based outcomes, which appear underpowered at the current sample size.

This matters substantively because the extension speaks most clearly to donation volume and much less clearly to downstream contracting counts. Mixed results in the count models should therefore be interpreted as limited evidence rather than as a sharp contradiction of the ideological-distance argument in either Acs (2025) or Fazekas et al. (2023).

Figure 4. Projected Power as the Matched Sample Size Grows

Figure 4 shows how uneven that power problem is. With the observed HW2 effect sizes, log donation is already above 0.80 at $N = 48$, total contracts reaches 0.80 only after the sample doubles to about 96, and donor-contract count does not cross 0.80 until the sample is much closer to the original ACS size. Under the half-effect scenario, even total contracts remains below 0.80 at $N = 276$, and donor-contract count stays far below that benchmark throughout. The figure therefore supports a cautious interpretation of the count-based procurement results: mixed or null findings there may reflect low power rather than strong evidence of no relationship.

5. Discussion

Implications

The HW3 robustness checks shift the interpretation of the ACS-Fazekas extension from simple sign reading to a more careful assessment of what the sample can actually support. The original ACS replication remains strong because it is estimated on a much larger agency-administration panel. The procurement extension is substantively interesting, but it is built on a matched sample that is small enough for wide uncertainty, case sensitivity, and low power to become central inferential concerns.

The bootstrap and leave-one-agency-out checks answer complementary questions. The bootstrap asks how uncertain the coefficient is under repeated sampling from a small matched panel. The leave-one-agency-out analysis asks whether the result is being carried by a handful of agencies. In this application, the two checks point to a mixed conclusion rather than a simple rejection of the extension. The donor-contract-count and total-contract models are directionally stable when agencies are omitted one at a time, but the power analysis shows that both are still underpowered at the current sample size. The log-donation model is the strongest case substantively, yet the bootstrap indicates that its interval is wider than the model-based standard errors imply. The donor-contract-share model does not display a stable signal under any of the three checks.

The power-style simulation adds the final piece. It shows whether the observed sample size is large enough to detect effects of the size reported in HW2. The answer depends on the outcome. The log-donation model is reasonably well powered if the observed coefficient is close to the true one. The count-based outcomes are not. That means null or mixed findings in those models are weak evidence against the theory. The better conclusion is that the extension contains one relatively informative outcome, one near-null outcome, and two outcomes for which low power remains a serious constraint.

In substantive terms, the clearest pattern in the matched panel is that agencies farther from the president are associated with lower total donation volume. The matched sample is too small, however, to say with the same confidence whether ideological distance also changes donor-contract incidence or contract counts.

Method-Theory Link

The theory in Acs (2025) is about ideological alignment between presidents and agencies. My HW2 extension translated that logic into procurement outcomes by asking whether the same ideological distance measure predicts donation and contracting behavior. HW3 does not introduce a new theory. Instead, it asks whether the extension is estimated precisely enough to support the substantive claims made from it.

This is the relevant method-theory link for the homework: when the theory is tested on a much smaller matched sample than the original article used, the main threat is not only omitted variables or model choice. It is also whether the data contain enough information to distinguish signal from noise. In Acs (2025), ideological distance is meant to capture how far agencies sit from presidential preferences. In Fazekas et al. (2023), political contributions are one channel through which firms try to shape procurement relationships. The bootstrap, leave-one-agency-out, and power-style simulation all address whether those two ideas can be connected credibly in a small matched panel. Together they make the extension more informative because they separate weak theory from weak information.

Limitations and Future Research

The main limitation of this HW3 design is that it treats the matched procurement panel as the object of inference without changing the underlying agency matching procedure. If the match itself is noisy, then the sample-size problem and the measurement problem are intertwined. A larger matched panel built from official agency identifiers would improve both coverage and statistical power.

A second limitation is that the power-style simulation is retrospective. It uses the observed HW2 estimates as anchors for plausible effect sizes. That is useful for diagnosing whether the current sample is informative, but it is not a substitute for a prospective design analysis based on a pre-specified substantive benchmark. A

potential next step would be to define a minimum substantively important effect on donation or contracting outcomes and use that benchmark for a prospective power analysis.

References

- Acs, A. (2025). [Mapping the political contours of the regulatory state: Dynamic estimates of agency ideal points](#). *American Political Science Review*, 1–18.
- Fazekas, M., Ferrali, R., & Wachs, J. (2023). [Agency independence, campaign contributions, and favoritism in US federal government contracting](#). *Journal of Public Administration Research and Theory*, 33(2), 262–278.